

**MACHINE LEARNING (Course Code: 25SC2107E)****Y25 - 2026-2027 Trimester - 4****Project Abstract Submission Form**

Sec No- 09	Team No- 23	Project Title- Match Outcome Prediction Using Logistic Regression and Random Forest
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1. Abstract :

Predicting the outcome of a sports match is an interesting application of machine learning because the result can depend on several factors such as team performance, recent form, venue, toss result, and previous match records. This project, "Match Outcome Prediction Using Logistic Regression and Random Forest," uses historical match data to predict the possible outcome of a future match. The data is first cleaned and prepared by handling missing values, selecting useful features, and performing feature engineering. Two supervised machine learning algorithms, Logistic Regression and Random Forest, are trained and compared to determine which model provides better predictions. Their performance is evaluated using accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC. The project can also include hyperparameter tuning and model deployment through a simple application or API, where users can enter match details and receive a predicted outcome with its probability. Overall, the project demonstrates how machine learning can be applied to real-world sports data while following the complete process from data preparation and model training to prediction and deployment.

2. Team Details :

Roll Number	Name of the Student	Signature
2520030027	Venkata Sai Krishna	
2520030320	Ajai Manikanta	

3. Frameworks / Platforms / Software Used :

The project uses Python as the main programming language, with Google Colab/Jupyter Notebook for development. Pandas and NumPy are used for data processing, while Scikit-learn is used to build the Logistic Regression and Random Forest models. Matplotlib and Seaborn are used for data visualization, and FastAPI is used for API development and model deployment. Postman is used for API testing, and Joblib is used to save and load the trained models.

Guide Name and Signature (remarks if any) _____